

Q1 - 24 June - Shift 1

The approximate height from the surface of earth at which the weight of the body becomes $\frac{1}{3}$ of its weight on the surface of earth is : [Radius of earth $R = 6400$ km and $\sqrt{3} = 1.732$]

- (A) 3840 km (B) 4685 km
(C) 2133 km (D) 4267 km

Space for your notes:

Q2 - 24 June - Shift 2

The distance between Sun and Earth is R . The duration of year if the distance between Sun and Earth becomes $3R$ will be :

- (A) $\sqrt{3}$ years (B) 3 years
(C) 9 years (D) $3\sqrt{3}$ years

Space for your notes:

Q3 - 25 June - Shift 1

The height of any point P above the surface of earth is equal to diameter of earth. The value of acceleration due to gravity at point P will be : (Given g = acceleration due to gravity at the surface of earth)

- (A) $g/2$
(B) $g/4$
(C) $g/3$
(D) $g/9$

Space for your notes:

Q4 - 25 June - Shift 2

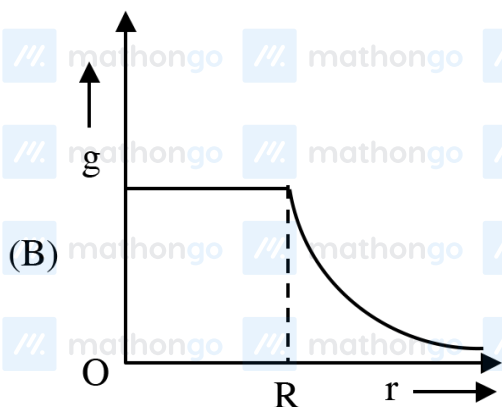
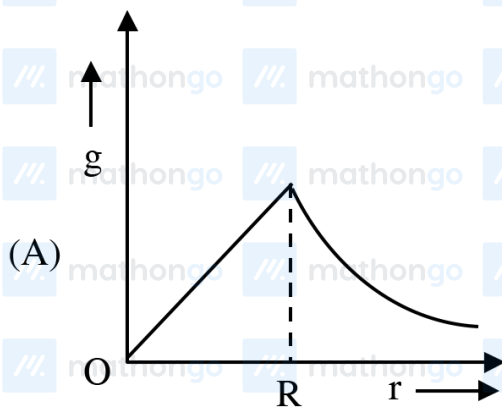
Questions

MathonGo

Two satellites S_1 and S_2 are revolving in circular orbits around a planet with radius $R_1 = 3200$ km and $R_2 = 800$ km respectively. The ratio of speed of satellite S_1 to the speed of satellite S_2 in their respective orbits would be $\frac{1}{x}$ where $x =$

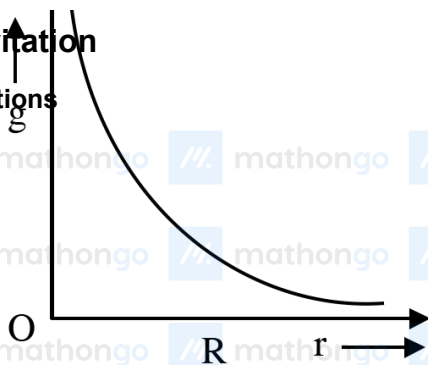
*Space for your notes:***Q5 - 26 June - Shift 1**

The variation of acceleration due to gravity (g) with distance (r) from the center of the earth is correctly represented by : (Given R = radius of earth)

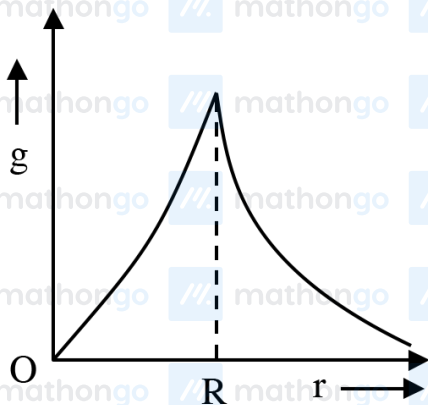
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(C)



(D)

**Q6 - 26 June - Shift 1**

The elongation of a wire on the surface of the earth is 10^{-4} m. The same wire of same dimensions is elongated by 6×10^{-5} m on another planet. The acceleration due to gravity on the planet will be ms^{-2} . (Take acceleration due to gravity on the surface of earth = 10 m/s^{-2})

*Space for your notes:***Q7 - 26 June - Shift 2**

Questions

MathonGo

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : If we move from poles to equator, the direction of acceleration due to gravity of earth always points towards the center of earth without any variation in its magnitude.

Reason R : At equator, the direction of acceleration due to the gravity is towards the center of earth.

In the light of above statements, choose the correct answer from the options given below :

- (A) Both A and R are true and R is the correct explanation of A.
- (B) Both A and R are true but R is NOT the correct explanation of A.
- (C) A is true but R is false
- (D) A is false but R is true

Space for your notes:

Q8 - 27 June - Shift 1

Given below are two statements :

Statement I : The law of gravitation holds good for any pair of bodies in the universe.

Statement II : The weight of any person becomes zero when the person is at the centre of the earth.

In the light of the above statements, choose the correct answer from the options given below.

- (A) Both statement I and Statement II are true
- (B) Both statement I and Statement II are false
- (C) Statement I is true but Statement II are false
- (D) Statement I is false but Statement II is true

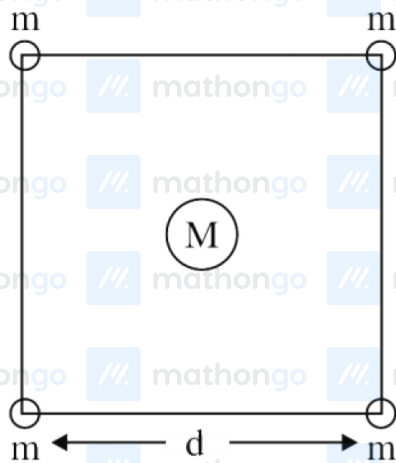
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Q9 - 27 June - Shift 2

Four spheres each of mass m form a square of side d (as shown in figure). A fifth sphere of mass M is situated at the centre of square. The total gravitational potential energy of the system is :

Space for your notes:



(A) $-\frac{Gm}{d}[(4 + \sqrt{2})m + 4\sqrt{2}M]$

(B) $-\frac{Gm}{d}[(4 + \sqrt{2})M + 4\sqrt{2}m]$

(C) $-\frac{Gm}{d}[3m^2 + 4\sqrt{2}M]$

(D) $-\frac{Gm}{d}[6m^2 + 4\sqrt{2}M]$

Q10 - 28 June - Shift 1

Questions

MathonGo

Two planets A and B of equal mass are having their period of revolutions T_A and T_B such that $T_A = 2T_B$. These planets are revolving in the circular orbits of radii r_A and r_B respectively.

Which out of the following would be the correct relationship of their orbits?

(A) $2r_A^2 = r_B^2$

(B) $r_A^3 = 2r_B^3$

(C) $r_A^3 = 3r_B^3$

(D) $T_A^2 - T_B^2 = \frac{\pi^2}{GM} (r_B^3 - 4r_A^3)$

Space for your notes:

Q11 - 28 June - Shift 2

Water fall from a 40 m high dam at the rate of 9×10^4 kg per hour. Fifty percentage of gravitational potential energy can be converted into electrical energy. Using this hydroelectric energy number of 100W lamps, that can be lit, is:

(Take $g = 10 \text{ ms}^{-2}$)

(A) 25

(B) 50

(C) 100

(D) 18

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Q12 - 28 June - Shift 2

Two objects of equal masses placed at certain distance from each other attracts each other with a force of F . If one-third mass of one object is transferred to the other object, then the new force will be :

- (A) $\frac{2}{9}F$ (B) $\frac{16}{9}F$
(C) $\frac{8}{9}F$ (D) F

Space for your notes:

Q13 - 29 June - Shift 1

The escape velocity of a body on a planet 'A' is 12 kms^{-1} . The escape velocity of the body on another planet 'B', whose density is four times and radius is half of the planet 'A', is :

- (A) 12 kms^{-1} (B) 24 kms^{-1}
(C) 36 kms^{-1} (D) 6 kms^{-1}

Space for your notes:

Q14 - 29 June - Shift 2

The time period of a satellite revolving around earth in a given orbit is 7 hours. If the radius of orbit is increased to three times its previous value, then approximate new time period of the satellite will be :

- (A) 40 hours (B) 36 hours
(C) 30 hours (D) 25 hours

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Answer Key

Q1 (B) Q2 (D) Q3 (D) Q4 (2)

Q5 (A) Q6 (6) Q7 (D) Q8 (A)

Q9 (A) Q10 (C) Q11 (B) Q12 (C)

Q13 (A) Q14 (B)

Q15 (A) Q16 (D) Q17 (D) Q18 (A)

Q19 (A) Q20 (D) Q21 (D) Q22 (A)

Q23 (A) Q24 (D) Q25 (D) Q26 (A)

Q27 (A) Q28 (D) Q29 (D) Q30 (A)

Q31 (A) Q32 (D) Q33 (D) Q34 (A)

Q35 (A) Q36 (D) Q37 (D) Q38 (A)

Q39 (A) Q40 (D) Q41 (D) Q42 (A)

Q43 (A) Q44 (D) Q45 (D) Q46 (A)

Q47 (A) Q48 (D) Q49 (D) Q50 (A)

Q51 (A) Q52 (D) Q53 (D) Q54 (A)

Q55 (A) Q56 (D) Q57 (D) Q58 (A)

Q59 (A) Q60 (D) Q61 (D) Q62 (A)

Q63 (A) Q64 (D) Q65 (D) Q66 (A)

Q67 (A) Q68 (D) Q69 (D) Q70 (A)

Q71 (A) Q72 (D) Q73 (D) Q74 (A)

Q75 (A) Q76 (D) Q77 (D) Q78 (A)

Q79 (A) Q80 (D) Q81 (D) Q82 (A)

Q83 (A) Q84 (D) Q85 (D) Q86 (A)

Q87 (A) Q88 (D) Q89 (D) Q90 (A)

Q91 (A) Q92 (D) Q93 (D) Q94 (A)

Q95 (A) Q96 (D) Q97 (D) Q98 (A)

Q99 (A) Q100 (D) Q101 (D) Q102 (A)

Q103 (A) Q104 (D) Q105 (D) Q106 (A)

Q107 (A) Q108 (D) Q109 (D) Q110 (A)

Q111 (A) Q112 (D) Q113 (D) Q114 (A)

Q115 (A) Q116 (D) Q117 (D) Q118 (A)

Q119 (A) Q120 (D) Q121 (D) Q122 (A)

Q123 (A) Q124 (D) Q125 (D) Q126 (A)

Q127 (A) Q128 (D) Q129 (D) Q130 (A)

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Q1 (B)

$$Mg' = \frac{M}{3}g$$

$$g' = \frac{g}{3}$$

$$g' = g \left(\frac{R}{R+h} \right)^2 = \frac{g}{3}$$

$$\frac{R}{R+h} = \frac{1}{\sqrt{3}}$$

$$h = (\sqrt{3} - 1)R$$

$$= (1.732 - 1)6400$$

$$\boxed{h = 4685 \text{ km}}$$

Q2 (D)

$$T' = T \left(\frac{3R}{R} \right)^{3/2} = 3\sqrt{3} T$$

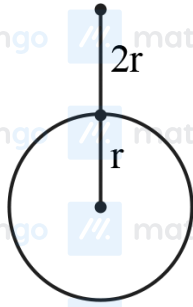
Q3 (D)

$$g = \frac{Gm}{r^2}$$

$$g' = \frac{Gm}{(3r)^2}$$

$$g' = \frac{Gm}{9r^2}$$

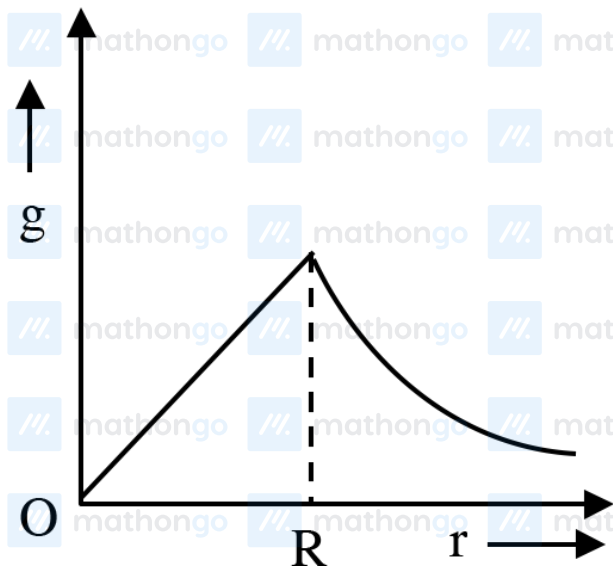
$$g' = \frac{g}{9}$$

**Q4 (2)**

$$V = \frac{GM}{r} \Rightarrow \frac{V_1}{V_2} = \sqrt{\frac{800}{3200}} = \frac{1}{2}$$

Q5 (A)

$$g = \begin{cases} \frac{GMr}{R^3}, & r \leq R \\ \frac{GM}{r^2}, & r \geq R \end{cases}$$



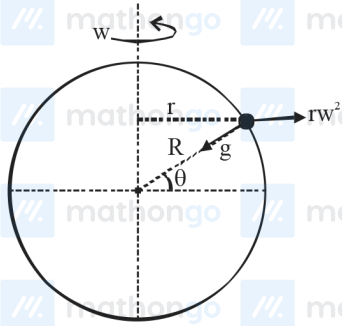
Q6 (6)

$$\Delta l \propto g$$

$$\frac{\Delta l_{\text{earth}}}{\Delta l_{\text{planet}}} = \frac{g_{\text{earth}}}{g_{\text{planet}}} = \frac{10^{-4}}{6 \times 10^{-5}}$$

$$g_{\text{planet}} = 6 \text{ m/s}^2$$

Q7 (D)



Effective acceleration due to gravity is the resultant of g & rw^2 whose direction & magnitude

depends upon θ . Hence assertion is false.

When $\theta = 0^\circ$ (at equator), effective acceleration is radially inward.

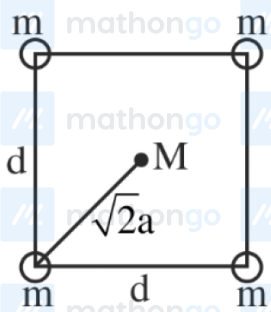
Q8 (A)

Since it is universal law so it hold good for any pair of bodies.

The value of g at centre is zero.

So statement I and Statement II are true.

Q9 (A)



$$-\frac{Gm^2}{d} \times 4 - \frac{Gm^2}{\sqrt{2}d} \times 2 - \frac{GMm}{d} \times 4\sqrt{2}$$

$$-\frac{Gm}{d} \left[(4 + \sqrt{2})m + 4\sqrt{2}M \right]$$

Q10 (C)

$$T = \frac{2\pi}{\sqrt{Gm_a}} r^{\frac{3}{2}}$$

$$T^2 \propto r^3$$

$$\left(\frac{T_A}{T_B}\right)^2 = \left(\frac{r_A}{r_B}\right)^3$$

$$\Rightarrow \left(\frac{2}{1}\right)^2 = \left(\frac{r_A}{r_B}\right)^3 \Rightarrow r_A^3 = 4r_B^3$$

Q11 (B)

$$\frac{9 \times 10^4 \times g \times 40}{3600} \times 0.5 = n \times 100$$

$$\frac{10^4 \times 0.5}{100} = n$$

$$100 \times 0.5 = n$$

$$n = 50$$

Q12 (C)

$$F = \frac{Gm^2}{r^2}$$

$$F' = \frac{G\left(\frac{4m}{3}\right) \times \left(\frac{2m}{3}\right)}{r^2}$$

$$F' = \frac{8}{9} F$$

Q13 (A)

$$V_{\text{escape}} = \sqrt{\frac{2Gm}{R}} \Rightarrow \sqrt{\frac{2G\rho \times \frac{4}{3}\pi R^3}{R}}$$

$$V_{\text{escape}} \propto \sqrt{\rho R^2}$$

$\therefore$ if ρ is 4 times and Radius is halved.

$\Rightarrow V_{\text{escape}}$ will remain same $\therefore$ Ans (A)

Q14 (B)

$$T = \frac{2\pi}{\sqrt{GM}} r^{3/2}$$

$$\frac{T_1}{T_2} = \left(\frac{r_1}{r_2}\right)^{3/2} = \left(\frac{1}{3}\right)^{3/2}$$

$$T_2 = T_1 \cdot 3\sqrt{3} = 21\sqrt{3} \text{ hours}$$
$$\approx 36 \text{ hours}$$